

ECON 320 – Introduction to Mathematical Economics**Quiz 1 | September 24, 2026**

University of Northern British Columbia Instructor: M. Karimzada

Part A: Multiple Choice (5 questions $\times$ 1 mark = 5 marks)

Q1. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 5, 6\}$, and $B = \{2, 3, 6, 7\}$. Which of the following correctly gives $(A \cap B)^c$?

- (A) $\{2, 6\}$
- (B) $\{1, 3, 4, 5, 7, 8\}$
- (C) $\{1, 2, 3, 5, 6, 7\}$
- (D) $\{4, 8\}$

Q2. For a set A with $|A| = 5$, the number of elements in the power set $\mathcal{P}(A)$ is:

- (A) 10
- (B) 25
- (C) 32
- (D) 64

Q3. Which of the following functions $f : \mathbb{R} \rightarrow \mathbb{R}$ is **injective but not surjective**?

- (A) $f(x) = x^2$
- (B) $f(x) = e^x$
- (C) $f(x) = x^3$
- (D) $f(x) = \sin x$

Q4. The inverse of $f(x) = \frac{2x+3}{x-1}$, $x \neq 1$, is:

- (A) $f^{-1}(x) = \frac{x+3}{x-2}$
- (B) $f^{-1}(x) = \frac{x-3}{x+2}$
- (C) $f^{-1}(x) = \frac{x-1}{2x+3}$

Name: _____

Student ID: _____

(D) $f^{-1}(x) = \frac{2x-1}{x+3}$

Q5. Let $f(x) = x^2 - 1$ and $g(x) = \sqrt{x+2}$. What is $(g \circ f)(3)$?

(A) $2\sqrt{2}$

(B) $\sqrt{10}$

(C) 4

(D) 3

Part B: Numerical Problems (15 marks)

Q6. (5 marks) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 3, 5, 7, 9\}$, $B = \{1, 2, 3, 4, 5\}$, and $C = \{3, 5, 6, 7, 8\}$.

(a) Find $A \cap B$ and $A \setminus B$. [2 marks]

(b) Find $(A \cup B)^c$. [1 mark]

(c) Verify De Morgan's Law: $(A \cup B)^c = A^c \cap B^c$. [1 mark]

(d) Determine $|\mathcal{P}(A \cap B \cap C)|$. [1 mark]

Q7. (5 marks) Let $f(x) = \frac{2x+3}{x-1}$, $x \neq 1$.

(a) Find the natural domain and range of f . [2 marks]

(b) Find $f^{-1}(x)$ and state its domain. [2 marks]

(c) Verify that $f(f^{-1}(x)) = x$. [1 mark]

Name: _____

Student ID: _____

Q8. (5 marks) A firm's profit function is $\pi(x) = -2x^2 + 12x - 10$, where $x \geq 0$ is the quantity produced.

(a) Find all quantities at which $\pi(x) = 0$ (the break-even outputs). [1 mark]

(b) Find the profit-maximising quantity x^* and the maximum profit π^* . [2 marks]

(c) Is π concave or convex? Justify using the second derivative. [1 mark]

(d) Find the range of π on $[0, \infty)$. [1 mark]

— End of Quiz — Total: 20 marks Time: 20 minutes Good luck!